

Instability and Damping in a One-Dimensional Gaussian Adjoint Sampler

1 Problem and terminology

We study a one-dimensional Gaussian-to-Gaussian problem with

$$p_{\text{tar}} = N(1, 0.02), \quad p_0 = N(1.05, 0.02), \quad \sigma(t) \equiv 1, \quad t \in [0, 1].$$

Here the *drift* is the deterministic part of the sampling SDE. An *outer iteration* means one cycle in which the current endpoint distribution is used to construct a new drift, after which the SDE is simulated again. In the learned cases, the drift is represented by a neural network. *Reciprocal adjoint matching* refers to the Adjoint Sampling training rule that constructs regression targets from current terminal samples and the target density. A *replay buffer* stores endpoint/gradient pairs from previous outer iterations and reuses them in later training. *Damping* means replacing a newly trained network parameter vector by a convex combination of the previous and proposed parameter vectors.

Because the target and initial variances are matched, the exact Gaussian outer map predicts

$$m_{k+1} - 1 = A(m_k - 1), \quad A = 1 + \frac{\log(0.02)}{1 - 0.02} \approx -2.99186.$$

Thus the exact mean error alternates in sign and grows by a factor of about three per outer iteration. The continuous prediction for the mean is

$$1.05, \quad 0.8504, \quad 1.4476, \quad -0.3390, \quad 5.0062.$$

The numerical experiments use 500 time steps with the RMC `q1` time-grid option. Learned runs use three random seeds. Histograms for learned regimes pool the final samples across all three seeds; consequently, their pooled variance reflects both within-seed spread and variation in the terminal mean across seeds.

Regime	Drift representation	How the drift is obtained	Replay	Damping
Affine oracle	Exact affine function	Closed-form Gaussian update from the current mean and variance	No	No
Learned affine	3×64 time-dependent neural network	Supervised regression onto the exact affine drift	No	No
Full Adjoint	Same neural network	Reciprocal adjoint matching from current terminal samples	1000	No
Full Adjoint + damping	Same neural network	Same as Full Adjoint	1000	$\eta = 0.5$

Table 1: The four regimes. In the damped case the first learned update is left undamped because p_0 is prescribed externally; parameter damping begins with the second outer update.

Quantity	Value	Quantity	Value
Training endpoints per outer step	128	Inner optimizer steps	100
Mini-batch size	64	Optimizer	Adam, 10^{-4}
Evaluation samples per seed	10,000	Oracle samples	50,000
Neural-network hidden widths	64, 64, 64	Time steps	500
Terminal-gradient clip	150	Learned seeds	0, 1, 2

Table 2: Shared numerical settings for the learned experiments.

2 Affine oracle

The oracle uses the exact affine Gaussian drift at every outer iteration, so it isolates the instability of the outer map from neural-network approximation and optimization error. The numerical mean trace is

$$1.0500, \quad 0.8490, \quad 1.4371, \quad -0.3122, \quad 4.8179,$$

which closely follows the continuous prediction. The variance remains near the target value: the final empirical variance is 0.02032.

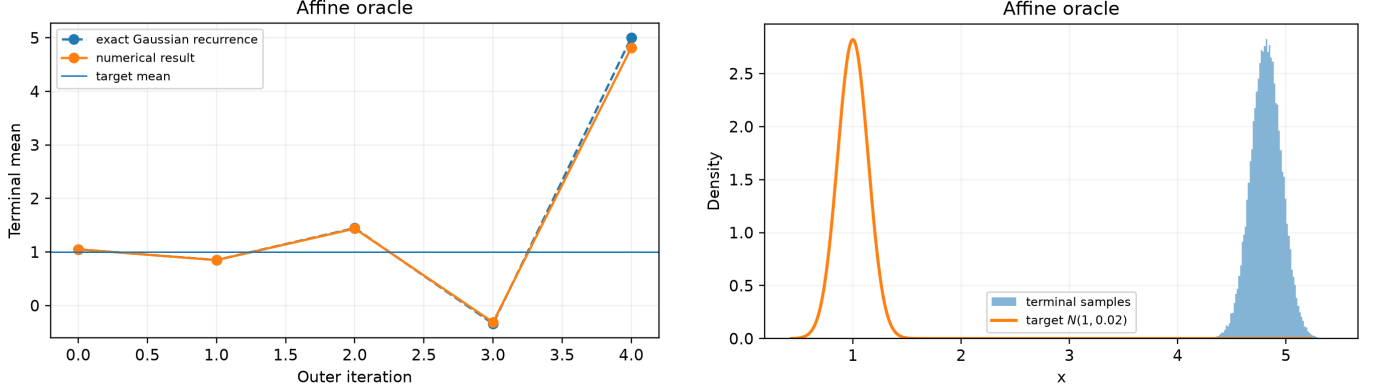


Figure 1: Affine oracle. Left: terminal mean across outer iterations. Right: final terminal samples with the exact $N(1, 0.02)$ density overlaid.

The oracle therefore gives the clean instability mechanism: the variance is essentially correct while the mean is driven away from the target by the alternating outer update.

3 Learned affine drift

This regime keeps the same neural-network architecture used by the practical sampler but trains it directly against the known affine drift. It removes the reciprocal-adjoint target construction and replay mechanism while retaining function approximation and finite optimization.

Averaged over the three seeds, the terminal mean follows

$$1.0500, \quad -0.3775, \quad 0.5597, \quad 2.5374, \quad 0.4758.$$

The final seed-to-seed standard deviation of the mean is 0.1263. The pooled final variance is 0.2069, more than ten times the target variance.

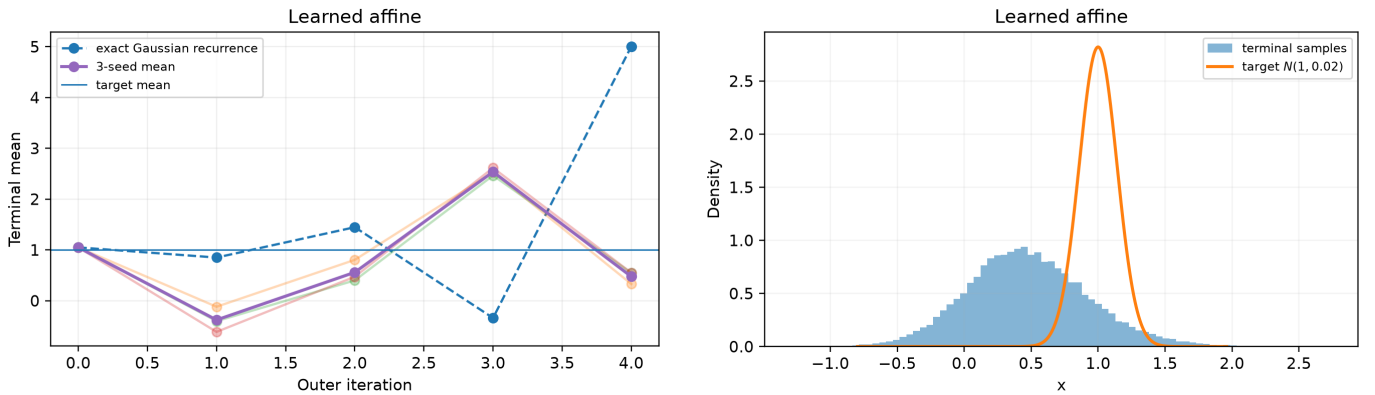


Figure 2: Learned affine drift. Thin curves show individual seeds and the heavier curve shows their mean. The histogram pools the final samples from all three seeds.

The mean remains strongly oscillatory, but the learned dynamics no longer follow the exact Gaussian recurrence. The variance inflation shows that fitting the drift with a finite neural network changes both the mean and contraction properties of the update.

4 Full Adjoint sampler

The full sampler uses reciprocal adjoint matching, rollout feedback, a 1000-sample replay buffer, and terminal-gradient clipping at 150. The mean averaged across seeds is

$$1.0500, \quad -0.3005, \quad 1.0064, \quad 2.1552, \quad 1.5859.$$

The final seed-to-seed standard deviation of the mean is 1.5032, substantially larger than in the learned-affine experiment. The three final means are 0.8464, 0.5957, and 3.3156. The pooled terminal variance is 1.6604.

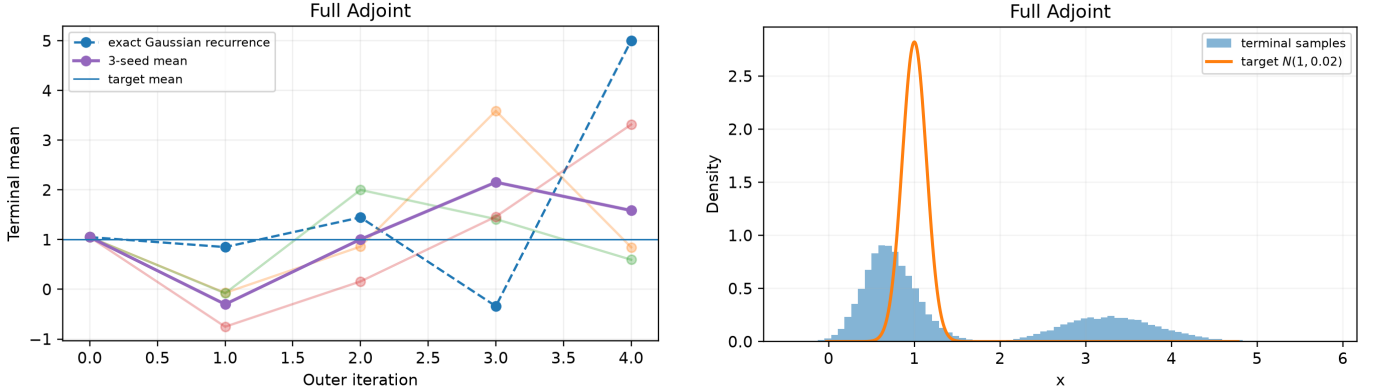


Figure 3: Full Adjoint sampler. The terminal distribution is broad and strongly seed-dependent after four outer iterations.

The practical training machinery substantially changes the exact outer-map dynamics. Instability remains visible through oscillation and seed sensitivity, but the dominant failure is now a combination of mean error and severe variance inflation. Terminal-gradient clipping is active in a few outer iterations, so the run also enters the safeguard regime of the implementation.

5 Full Adjoint sampler with damping

The damped sampler is identical to the full sampler except that, after each inner training block from the second outer iteration onward, the proposed neural-network parameters $\hat{\theta}_{k+1}$ are replaced by

$$\theta_{k+1} = (1 - \eta)\theta_k + \eta\hat{\theta}_{k+1}, \quad \eta = 0.5.$$

The first update is intentionally undamped, so the first-step results agree exactly with the undamped sampler. If the same relaxation acted directly on the scalar mean error, the ideal multiplier would be

$$1 - \eta + \eta A \approx -0.99593,$$

which is only barely contractive. Parameter averaging in a nonlinear network is not identical to scalar mean relaxation, so this value serves only as a reference.

The three-seed mean trace is

$$1.0500, \quad -0.3005, \quad 0.1383, \quad 0.8518, \quad 1.6319.$$

The final seed-to-seed standard deviation of the mean falls from 1.5032 without damping to 0.8402 with damping. The pooled terminal variance falls from 1.6604 to 0.8487, although it remains far above the target variance 0.02.

6 Comparison

The experiment separates three effects. First, the exact Gaussian update has a genuine outer-iteration instability even while preserving the target variance accurately. Second, finite neural-network fitting substantially modifies that ideal map and introduces variance error. Third, the full reciprocal-adjoint/replay implementation is considerably more seed-sensitive than the direct affine regression experiment. Parameter damping reduces that seed sensitivity and cuts the pooled terminal variance by roughly one half, but $\eta = 0.5$ is near the scalar stability boundary for this target variance and does not yield convergence over the four iterations tested. These results support damping as a stabilizing modification while also showing that the practical sampler has approximation and variance-control errors beyond the one-dimensional mean instability predicted by the Gaussian analysis.

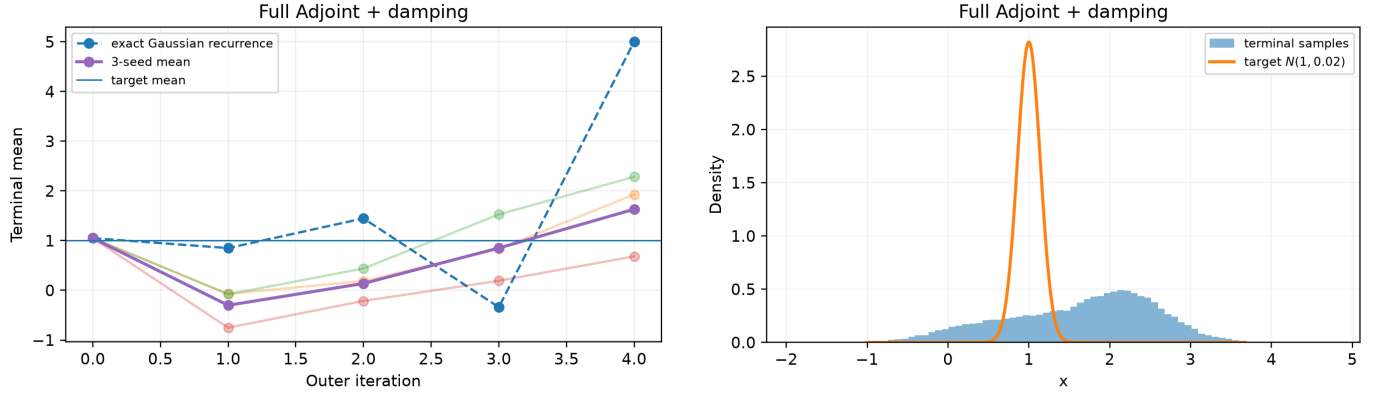


Figure 4: Full Adjoint sampler with parameter damping $\eta = 0.5$. The dashed curve in the mean panel is the undamped exact Gaussian recurrence and is shown only as a reference. Damping reduces seed-to-seed spread and terminal variance but does not recover the target distribution within four outer iterations.

Regime	Final pooled mean	Final pooled variance	SD of final seed means
Affine oracle	4.8173	0.0203	—
Learned affine	0.4758	0.2069	0.1263
Full Adjoint	1.5859	1.6604	1.5032
Full Adjoint + damping	1.6319	0.8487	0.8402

Table 3: Terminal comparison after four outer iterations. The learned pooled statistics combine all three seeds.